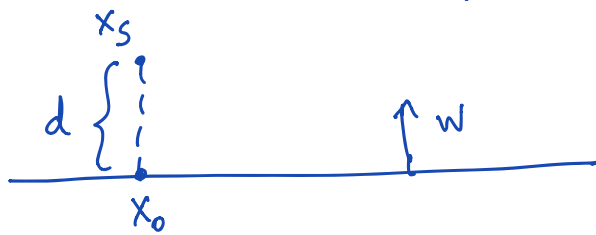


SVM : calcul distance entre un point x_s et l'hyperplan H (en $\mathbb{R}^n$)



$$H: w \cdot x + b = 0$$

$$w \cdot x = \langle w, x \rangle = w^T \cdot x$$

(produit scalaire)

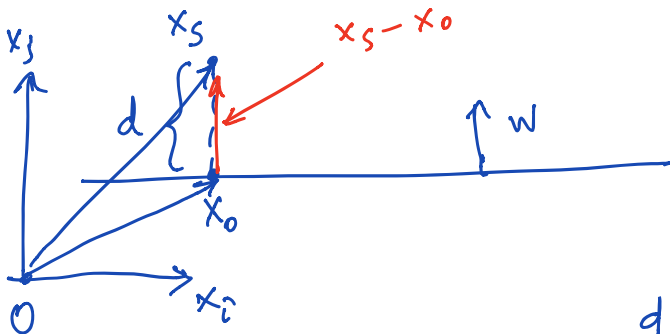
$$w = (w_1, w_2, \dots, w_n)$$

$$x_s = (x_{s0}, x_{s1}, \dots, x_{sn})$$

$$x = (x_1, x_2, \dots, x_n)$$

$$w \cdot x = w^T \cdot x = [w_1 \ w_2 \ \dots \ w_n] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} =$$

$$= w_1 x_1 + w_2 x_2 + \dots + w_n x_n$$



$$x_s - x_0 \parallel w$$

$$d(x_s, H) = d(x_s, x_0) = \|x_s - x_0\| = d$$

$$d \rightarrow \frac{x_s - x_0}{\|x_s - x_0\|} = \frac{w}{\|w\|} \quad (\text{norme} = 1)$$

$$\frac{x_s - x_0}{d} = \frac{w}{\|w\|} \quad \Bigg| \cdot w$$

$$\frac{w \cdot (x_s - x_0)}{d} = \frac{w \cdot w}{\|w\|} = \frac{\|w\|^2}{\|w\|} = \|w\|$$

$$d \cdot \|w\| = w \cdot (x_s - x_0) = w \cdot x_s - w \cdot x_0 = w \cdot x_s \overset{=0}{\underbrace{- w \cdot x_0 - b}} + b =$$

$x_0 \in H$

$$= w \cdot x_s + b$$

$$d \cdot \|w\| = w \cdot x_s + b \Rightarrow \boxed{d = \frac{|w \cdot x_s + b|}{\|w\|}}$$